

Business Requirements Document (BRD)

1. Project Overview

Project Name: AI CHATBOT PROJECT

Version: 2.0

Date: \$(date)

Description: Modern full-stack AI assistant web app with dual interfaces:

- Chat Mode** (ChatGPT-style): Real-time conversations using Llama 3.1-8B on Cerebras.ai.
- Blackbox/Terminal Mode**: Developer terminal for code editing, file ops, command execution (AI-powered software engineering).

React frontend + FastAPI backend. Production-ready with animations, responsive design, PDF export.

2. Business Objectives

- Deliver fast, free AI chat & code assistance.
- Showcase Cerebras.ai inference (Llama 3.1).
- Enable seamless FE-BE integration for AI tools.
- **Success Metrics:**

Metric	Target
Response Time	<3s avg
UI Performance	60fps
User Sessions	Demo: 100% success
Export Usage	PDF gen works

3. Stakeholders

Role	Responsibilities
End Users	AI chat, code tasks
Developers	Extend Blackbox tools
Admins	Manage CEREBRAS_API_KEY
AI Provider	Cerebras.ai free tier

4. Functional Requirements

FR-1: Chat Interface

- Sidebar: BLACKBOX AI header, Chat/Terminal nav, New Chat/Export.
- Messages: User left (👤), Bot right (🤖), markdown/code rendering.
- Input: Multiline textarea (Shift+Enter new line), send (Enter).
- Features: Auto-scroll, typing indicator, skeletons, history clear, PDF export.

FR-2: Backend API

- `GET /: Health {"message": "Backend running"}`
- `POST /ask {question: str}` → Cerebras Llama response.

User Stories

- As user, I send query → get AI answer with markdown.
- As user, I export chat → PDF download.

5. Non-Functional Requirements

- | Category | Requirement |
- | **Performance** | <5s responses, smooth 60fps animations |
- | **Scalability** | Local dev, ready for cloud |
- | **Security** | CORS *, API key in .env |
- | **Accessibility** | Keyboard nav, ARIA, mobile responsive |
- | **Browsers** | Chrome/Edge/Firefox latest |
- | **Tech** | PWA-ready |

6. Technical Architecture

Browser (localhost:5173)

↓ HTTP POST /ask

FastAPI Backend (localhost:8000)

↓ OpenAI Client

Cerebras.ai (llama3.1-8b)

****Tech Stack**:**

Frontend:

└─ React 18 + Vite + TailwindCSS/Framer Motion

└─ React Router + React Markdown + html2pdf.js

Backend:

└─ FastAPI + Uvicorn + OpenAI SDK

└─ Pydantic + python-dotenv

AI: Cerebras Llama 3.1-8B

8. Acceptance Criteria

- [x] Chat: Query → Llama response <5s.
- [x] UI: Glassmorphism, animations, mobile responsive.
- [x] Nav: Route switching works.
- [x] Blackbox: Tools mock UI rendered.
- [x] Export: PDF generates correctly.
- [x] Error Handling: Friendly messages.

9. Risks & Dependencies

| Risk | Mitigation |

| ****High**** | CEREBRAS_API_KEY → Fallback mock responses |

| ****Medium**** | Blackbox incomplete → Phase 2: Real tools integration |

| ****Low**** | CORS/Ports → Configured |